

19	$\tan \theta = 0.75 / 1.2$ $\theta = 32.005^\circ$ $d \sin \theta = n \lambda$ $d = (2)(500 \times 10^{-9}) / \sin (32.005^\circ)$ $= 1.887 \times 10^{-3} \text{ mm}$ $d^{-1} = 529.9$ $= 530 \text{ lines per mm}$
20	Fundamental: $N A \rightarrow \lambda = 4L \rightarrow f = 80 \text{ Hz}$ 1st overtone: $N A N A \rightarrow \lambda = 4/2 L \rightarrow f = 2 \times 80 \text{ Hz} = 160 \text{ Hz}$